CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Simplifying $5(x+2)-3(x-4)$ gives: 1

(a) $2x-2$ **(b)** $8x+22$ **(c)** $2x+14$ **(d)** $2x+22$

2. 36 km/h expressed in m/s is: 1

(a) 12 m/s **(b)** 129.6 m/s **(c)** 0.6 m/s **(d)** 10 m/s

3. Two successive discounts of 10% and 10% are given on a marked price ₹ 2000. The selling price (in ₹) is: 1

(a) 1458 **(b)** 1782 **(c)** 1600 **(d)** 1620

4. An isosceles triangle has equal sides 39 cm and base 30 cm. Its height (in cm) is: 1

(a) 35 **(b)** 37 **(c)** 36 **(d)** 24

5. In a pie chart of 144 students a sector has a central angle of 80°. The number of students it represents is: 1

(a) 64 **(b)** 8 **(c)** 35 **(d)** 32

6. $(4x+1)(3x+2)$ equals: 1

(a) $12x^2-11x+2$ **(b)** $12x^2+11x-2$ **(c)** $12x^2+11x+2$ **(d)** $12x^2+2$

7. Two numbers are in the ratio 5 : 8 and their sum is 78. The larger number is: 1

(a) 54 **(b)** 30 **(c)** 48 **(d)** 39

8. An article is sold for ₹ 2400 at a profit of 20%. Its cost price (in ₹) is: 1

(a) 2000 **(b)** 1920 **(c)** 1800 **(d)** 2200

9. A rectangle has a diagonal of 39 cm and one side 15 cm. Its perimeter (in cm) is: 1

(a) 102 **(b)** 51 **(c)** 104 **(d)** 108

10. If A:B = 2:7 and B:C = 14:5, then A:C is: 1

(a) 2:5 **(b)** 10:98 **(c)** 2:3 **(d)** 4:5

11. $(3x-1y)^2$ equals: 1

(a) $9x^2-1y^2$ **(b)** $9x^2-6xy+1y^2$ **(c)** $9x^2-3xy+1y^2$ **(d)** $9x^2+6xy+1y^2$

12. Two numbers are in the ratio 2 : 7. If 3 is added to each, the numbers become 15 and 45. The original smaller number is: 1

(a) 15 **(b)** 24 **(c)** 9 **(d)** 12

13. A watch is sold for ₹ 2800 at a loss of 20%. Its cost price (in ₹) is: 1

(a) 3500 **(b)** 3150 **(c)** 3360 **(d)** 3850

14. The side of a rhombus is 39 cm and one diagonal is 30 cm. The other diagonal (in cm) is: 1

(a) 74 **(b)** 48 **(c)** 36 **(d)** 72

15. 9 workers working 6 hours a day finish a job in 20 days. 15 workers working 4 hours a day will finish it in (days): 1

(a) 12 **(b)** 20 **(c)** 180 **(d)** 18

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Simplify $9(x+3)-4(x-2)$. 2

17. A map has scale 1 : 50000. Two places are 6 cm apart on the map. Find the actual distance in km. 2

18. The price of a book rises from ₹ 160 to ₹ 200. Find the percentage increase. 2

19. The side of a rhombus is 50 cm and one diagonal is 60 cm. Find the other diagonal. 2

20. A pie chart of 108 students has a sector of 120°. How many students does it represent, and what percent of the students is that? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. In a pie chart, the sector for Maths measures 60° and represents 15 students. Find the total number of students and the number of students represented by a 96° sector. 3

22. Expand $(4x+1)(3x-2)$. 3

23. A mixture of 60 L contains milk and water in the ratio 3 : 2. How much water must be added to make the ratio 3 : 4? 3

24. A shopkeeper allows successive discounts of 20% and 10% on a marked price of ₹ 2500. Find the selling price and the single equivalent discount percent. 3

OR

A ladder 25 m long reaches a point 20 m up a wall. The foot is then pulled away so that it is 20 m from the wall. How far down does the top of the ladder slip? 3

25. 15 workers working 8 hours a day can finish a job in 12 days. In how many days can 18 workers working 5 hours a day finish it? 3

OR

Find 108^2 using an algebraic identity. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. An isosceles triangle has equal sides of 40 cm and base 48 cm. Find its height, area and perimeter. 5

OR

12 workers can complete a task in 20 days. After 8 days, 6 more workers join them. In how many more days will the work be finished? 5

27. A square plot has side $(4x + 5)$ m. A square pond of side $2x$ m is dug inside it. Write the area of the remaining land as an expression in x and find it when $x = 10$. 5

OR

20 men can complete a piece of work in 24 days. 5

(i) How many days will 16 men take?

(ii) How many men are needed to finish it in 30 days?

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND